

Object-Oriented Programming (OOP, CCIT4023)

HKU SPACE Community College, 2022-2023

Assignment 1

(15%)

(Total Marks: 30)

- Finish this work, based on concepts and techniques learnt in our course.
- Students should finish reviewing the related course notes and materials, before doing this assignment.
- Individual work: Student MUST FINISH THIS WORK ALONE. Student cannot work with others.
* *Plagiarism / Collusion / Shared work with others are not allowed. Zero mark will be given, with possible disciplinary action.*
- Student may reference and reuse the given samples and codes in our course materials, unless specified.
- Follow given instructions and guidelines.

Section A, A1A (10 marks): ONLINE (SOUL-Quiz Feature)

Multiple Choice (MC) and Matching Questions

- Identify and select the option of choice that "best" completes the statement, matches the item, or answers the question.
- **Number of Attempts Allowed: 2**
- **Grading method: Highest Grade**
- **Make sure you have *successfully* completed, "Finished" and submitted before deadline.**
*** Attempts must be submitted before time expires, or they are NOT counted (ZERO mark given).**
- Questions related to program codes are based on Java programming language, unless specified.
- 10 MC questions and 10 Matching questions. Each question carries the same mark.

Section B, A1B (20 marks): Programming Tasks

** Students may reference to the Practicing Labs to complete this task. Refer to the lab documents for details if necessary.*

General requirements (unless further specified):

- Students should follow what we have learnt in the course materials, including the lab materials.
- Unless specified, *no data validation required* in the program (program runs in normal situations and user input data as expected).
- Unless specified, all related files (including *.java, *.class) should be working *in the same folder*.
- Unless specified, do not create your own package.
- Each source code *.java file must *start with comments*, including the information of individual student (*student name, student id, class number, year of study*) as example below, unless specified.

```
// A0B1.java, For A0, OOP, 2022
/**
 * @author Student: CHAN Tai Man, 21124025, CL02, 2022
 */
public class A0B1{
```

Given Materials:

- This assignment document.
- Java files **A1B1.java** and **A1B2.java**: to be modified and completed by student. Also *modify the top comments for your own Student Information.*

A1B1 [10 marks]

1. Write a Java **program with user options** (with a class **A1B1** defined in a file named **A1B1.java**), by using and modifying the given file **A1B1.java**
 - Write and complete the following static methods in the program **A1B1.java**:
 1. **public static void showWelcome ()** :
 - display “Welcome!” on the console
 2. **public static int[] getRaNum ()** :
 - return **5 integers** which are generated **randomly from 21 to 98**
 3. **public static void dispMenu ()** :
 - display a menu which prompts user to select an option (Please refer to the *Sample Program Output* on the next page)
 4. **public static void main(String[] args)** :
 - the program entry point
 - The program runs continuously with loops, until exit by user. Each loop should:
 - Display a menu (by calling the method **dispMenu ()**)
 - Ask user for an option (0, 1, 2, 3, or other integers)
 - Based on the user input option, do the corresponding job, including *properly calling the methods* in the given class.

User Option (integer)	Response Action (Please refer to the <i>Sample Program Output</i> on the next page)
0	Exit the program
1	Display “Welcome!” by calling the method <code>showWelcome ()</code>
2	Call the method <code>getCurTime ()</code> , and show current time on the console
3	Call the method <code>getRaNum ()</code> , and show the array of random numbers on the console
All others	Display an ‘unknown’ message as shown in the <i>Sample Program Output</i>

- For console display of the array of random numbers, a reference code to convert an integer array (**numArray**) to a string is given below:

```
System.out.println( " Random numbers: " + java.util.Arrays.toString(numArray) );
```

Notes:

* Program must start with a proper message of **title**, as sample program output.

* Program must end with a proper message including **student information**, as sample program output.

Sample Program Output

```
--- START of A1B1 ---

>> Options (Integer only):
1: Say Welcome
2: Check Time
3: Random Numbers
0: Exit the program
.. Select an options: 1 <----- User Input
Welcome!

>> Options (Integer only):
1: Say Welcome
2: Check Time
3: Random Numbers
0: Exit the program
.. Select an options: 2
Current Time is 12:28:32

>> Options (Integer only):
1: Say Welcome
2: Check Time
3: Random Numbers
0: Exit the program
.. Select an options: 3
Random numbers: [64, 35, 26, 92, 47]

>> Options (Integer only):
1: Say Welcome
2: Check Time
3: Random Numbers
0: Exit the program
.. Select an options: 4
UNKNOWN option <4>.. try again!

>> Options (Integer only):
1: Say Welcome
2: Check Time
3: Random Numbers
0: Exit the program
.. Select an options: 0

>>> END of main(). Done by CHAN Tai Man, 21124025, CL02, 2022 <<<
```

A1B2 [10 marks]

1. Write a Java **program of Guessing Game** (with a class **A1B2** defined in a file named **A1B2.java**).
 - This guessing game lets a player guess a “randomly-picked integer number” **ranNum**, within a range (**min~max**).
 - Use **123** in your program, as the randomly-picked integer number **ranNum**.
 - Use the **last digit of your SID** (student ID) as **min**
 - Use the **first three digits of your SID** (student ID) as **max**
 - e.g. Use **5** as **min** and **211** as **max**, if the SID is **21124025**
 - Maximum number of attempts is set with **8** attempts. Quit with a proper message if maximum attempts reached, as the sample program output.
 - Each guess of the player will be followed with a feedback message to the player based on the guessed number and the randomly picked number, until the guessed number matched the picked number.
 - The program should interact with the user with console, as the sample program output.

Notes:

* Program starts with a proper message of **title**, as sample program output.

* Program ends with a proper message including **student information**, as sample program output.

Sample Program Output

--- START of **A1B2**, The Guessing Game ---

Guess a number between **5 to 211**, max number of attempts is 8

Attempt 1: Guess a random number <between 5 to 211>, then <ENTER>: 15

.. The number you typed is [15] - Too SMALL, guess the number again!

Attempt 2: Guess a random number <between 5 to 211>, then <ENTER>: 21

.. The number you typed is [21] - Too SMALL, guess the number again!

Attempt 3: Guess a random number <between 5 to 211>, then <ENTER>: 33

.. The number you typed is [33] - Too SMALL, guess the number again!

Attempt 4: Guess a random number <between 5 to 211>, then <ENTER>: 46

.. The number you typed is [46] - Too SMALL, guess the number again!

Attempt 5: Guess a random number <between 5 to 211>, then <ENTER>: 56

.. The number you typed is [56] - Too SMALL, guess the number again!

Attempt 6: Guess a random number <between 5 to 211>, then <ENTER>: 66

.. The number you typed is [66] - Too SMALL, guess the number again!

Attempt 7: Guess a random number <between 5 to 211>, then <ENTER>: 73

.. The number you typed is [73] - Too SMALL, guess the number again!

Attempt 8: Guess a random number <between 5 to 211>, then <ENTER>: 90

.. The number you typed is [90] - Too SMALL, guess the number again!

.. FAIL! Max Attempt reached! Our random number is [123]... QUIT!

>>> END of main(). Done by **CHAN Tai Man, 21124025, CL02, 2022** <<<

User
Input

↓

--- START of A1B2, The Guessing Game ---

Guess a number between 5 to 211, max number of attempts is 8

Attempt 1: Guess a random number <between 5 to 211>, then <ENTER>: 50

.. The number you typed is [50] - Too SMALL, guess the number again!

Attempt 2: Guess a random number <between 5 to 211>, then <ENTER>: 168

.. The number you typed is [168] - Too LARGE, guess the number again!

Attempt 3: Guess a random number <between 5 to 211>, then <ENTER>: 80

.. The number you typed is [80] - Too SMALL, guess the number again!

Attempt 4: Guess a random number <between 5 to 211>, then <ENTER>: 93

.. The number you typed is [93] - Too SMALL, guess the number again!

Attempt 5: Guess a random number <between 5 to 211>, then <ENTER>: 137

.. The number you typed is [137] - Too LARGE, guess the number again!

Attempt 6: Guess a random number <between 5 to 211>, then <ENTER>: 123

.. WELL DONE! This is the random number [123]... QUIT!

>>> END of main(). Done by CHAN Tai Man, 21124025, CL02, 2022 <<<

Evaluation and Submission

- Mark sure students follow instructions, including to follow the required *naming (e.g. parameters, methods and files)*.
- Compile, Run, Debug, Test and Evaluate your program based on the requirements.
- Submit ALL required files to SOUL:
 - A1B1.java, A1B1.class
 - A1B2.java, A1B2.class
- **Do NOT compress/zip or rename** the files. Submission work not following instructions and requirements may not be marked or be penalized.

* Remarks:

- *More testing may be performed during assessment based on the requirements, apart from the given testing cases.*

~ END ~